

Human-Made Diagram Layouts Are Not Minima of Edge-Length or Stress Energies

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Abstract

A layout tool places the boxes of a diagram by minimising a weighted sum of aesthetic criteria, and learning the weights from drawings people made has been proposed since 1994. Fitting the weights to the coordinates, as inverse optimisation does, assumes each drawing is a local minimum of some weighting. We test that one criterion at a time: move each box a fixed distance in sixteen directions, the rest fixed, and call the box held if no move lowers the criterion; q is the fraction of boxes held. A converged layout must read near 1 or the test lacks power for that criterion, and a random layout says what held costs by chance.

On 853 working diagrams of 15 to 40 boxes from three communities, laid out in their editors by the people who made them (hand layouts below), overlap holds every box of the median drawing at value zero. Uniform edge length and stress hold 0.00 of the median drawing there, in 136 laboratory drawings and in 1,072 figures from this field's proceedings, while a stress minimiser's own layouts are held at 1.00, and the hand layouts' stress is five to twelve times the minimiser's; a weight of 0.001 on either term beside crossings and overlap holds under 0.015 of a corpus's boxes, so no weighting of the base terms with positive weight on a distance term holds a hand layout. Offered the base energy, inverse optimisation returns all weight on overlap or crossings, the terms every layout satisfies. What the coordinates do hold is alignment into rows and columns: 58, 36 and 81 per cent of the median drawing's boxes lie in an alignment of three or more, none of the stress minimiser's, and a pairwise alignment term holds 0.52, 0.21 and 0.85 of them against 0.06 to 0.07 by chance and 0.06 to 0.09 after snapping a random layout to the same grid, the count tracking the share of boxes that share an exact coordinate with another, 0.54, 0.17 and 0.87. Neither base energy, Davidson and Harel's sum or stress, prices it; confined to the terms no layout satisfies exactly, the fit puts 0.88 to 0.95 of its weight on alignment with a stable held-out hold. Whether the person aligned the boxes or the editor's guide did, the coordinates cannot say, and for fitting it does not matter. Offer alignment among the terms before any fit, and judge a fitted energy by the fraction of a held-out drawing's boxes it holds under the terms no layout satisfies exactly.

1 Introduction

A layout program places the boxes of a diagram by minimising an *energy*, a weighted sum of quantities the field has agreed are bad [1, 2, 3, 4], and since 1994 the proposal has been to learn the weights from layouts people made or preferred [5, 6, 7, 8, 9]. Learning from preferences needs only that an accepted layout rank well under the learned energy; fitting to the coordinates, which is what inverse optimisation does [10, 11, 12], what O'Donovan, Agarwala and Hertzmann did for single-page designs [13], and what Flud [8] rests on when it continues a descent from crowd layouts, needs the coordinates themselves to sit at a local minimum. In graph drawing that property has not been tested, 4,890 figures were just collected from this field's proceedings into

*Independent researcher. Code, parsed corpora and every table's script: <https://github.com/Anode1/cjitter>, directory `example/diagrams`.

corpus	term		hand	neato	dot	ELK	random
WikiPathways	crossings	1.00 [1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.029)	1.00 (0.036)	0.67
	overlap	1.00 [1.00, 1.00]	(0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)	0.67
	length	0.00 [0.00, 0.00]	(0.219)	0.24 (0.011)	0.00 (0.241)	0.00 (0.127)	0.00 (0.193)
	stress	0.00 [0.00, 0.00]	(0.217)	1.00 (0.042)	0.00 (0.201)	0.00 (0.241)	0.00 (0.435)
	orthogonality	0.19 [0.18, 0.21]	(0.157)	0.00 (0.280)	0.19 (0.244)	0.19 (0.158)	0.00
	alignment A1	0.52 [0.48, 0.54]	(0.003)	0.06 (0.011)	1.00 (0.000)	0.57 (0.001)	0.07
	gridiness	0.87 [0.84, 0.89]	(0.424)	0.78 (1.000)	1.00 (0.125)	0.76 (0.200)	
	node-edge	0.88 [0.86, 0.90]	(0.004)	0.89 (0.000)	0.78 (0.011)	0.74 (0.016)	0.12
Reactome	flow	0.59 [0.56, 0.62]	(0.131)	0.50 (0.211)	1.00 (0.000)	1.00 (0.000)	0.41
	crossings	1.00 [1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.056)	1.00 (0.066)	0.61
	overlap	1.00 [1.00, 1.00]	(0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)	0.67
	length	0.00 [0.00, 0.00]	(0.234)	0.35 (0.005)	0.00 (0.232)	0.00 (0.205)	0.00 (0.188)
	stress	0.00 [0.00, 0.00]	(0.212)	1.00 (0.034)	0.00 (0.177)	0.00 (0.197)	0.00 (0.469)
	orthogonality	0.17 [0.15, 0.18]	(0.193)	0.00 (0.277)	0.15 (0.256)	0.12 (0.203)	0.00
	alignment A1	0.21 [0.19, 0.24]	(0.004)	0.06 (0.009)	1.00 (0.000)	0.42 (0.002)	0.06
	gridiness	0.65 [0.62, 0.68]	(0.640)	0.77 (1.000)	0.95 (0.176)	0.74 (0.306)	
BPMN	node-edge	1.00 [0.94, 1.00]	(0.000)	0.93 (0.000)	0.80 (0.007)	0.75 (0.021)	0.11
	flow	0.62 [0.60, 0.67]	(0.096)	0.45 (0.217)	1.00 (0.000)	1.00 (0.000)	0.36
	crossings	1.00 [1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)	0.62
	overlap	1.00 [1.00, 1.00]	(0.000)	0.41 (0.071)	1.00 (0.000)	1.00 (0.000)	0.60
	length	0.00 [0.00, 0.00]	(0.284)	0.40 (0.005)	0.00 (0.222)	0.00 (0.250)	0.00 (0.215)
	stress	0.00 [0.00, 0.00]	(0.197)	1.00 (0.016)	0.00 (0.152)	0.00 (0.175)	0.00 (0.507)
	orthogonality	0.73 [0.71, 0.75]	(0.011)	0.00 (0.277)	0.27 (0.169)	0.25 (0.145)	0.00
	alignment A1	0.85 [0.82, 0.89]	(0.001)	0.06 (0.011)	1.00 (0.000)	0.59 (0.001)	0.06
	gridiness	0.95 [0.94, 0.96]	(0.188)	0.79 (1.000)	0.81 (0.238)	0.67 (0.279)	
	node-edge	1.00 [1.00, 1.00]	(0.000)	1.00 (0.000)	0.86 (0.001)	0.79 (0.011)	0.11
	flow	0.86 [0.83, 0.88]	(0.041)	0.62 (0.120)	1.00 (0.000)	1.00 (0.000)	0.28

Table 1: $q_{0.02}$, the median over diagrams of the fraction of boxes no move of 0.02 of the width in sixteen directions improves, per criterion alone, on the hand layout with its 95% bootstrap interval, on **neato**, **dot** and ELK layouts of the same graphs, and on a uniform random layout; in parentheses the median value of the term at the layout, 0 when satisfied. Terms and controls are defined in Section 2, the corpora in Section 3.

one corpus [14], and this paper is the check to run before the first fit: move each box of a drawing a small distance in sixteen directions, call it *held* if no move lowers a criterion, and report the fraction of boxes held, q , per criterion, against a converged layout (which must read near 1) and a random one (chance). Table 1 is the result: no distance term holds a hand-placed box, no reweighting repairs it, and what the coordinates hold is an alignment into rows and columns that the base energies omit and that the coordinates cannot attribute to the hand rather than the editor’s guide. Section 2 is the test and what it rests on, Section 3 the corpora, Section 4 the measurements, Section 5 what else could produce them, Section 6 what was known and Section 7 what follows for a tool that learns its weights; the sensitivities are in a supplement and every table’s script in the repository.

2 The test

The energy. A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes’ sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^K w_k T_k(x), \quad w_k \geq 0, \quad (1)$$

where each term T_k is a quantity to be made small. Multicriteria layout is the choice of the T_k and the w_k [2, 3].

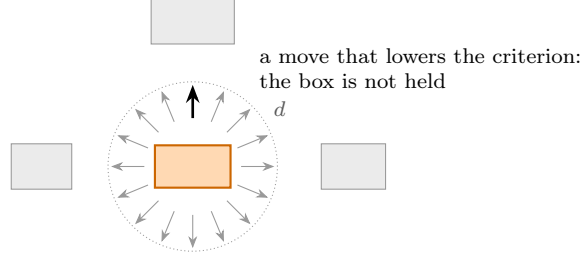


Figure 1: The test, one box and one criterion. The box is moved a distance d , two per cent of the drawing’s width, in each of sixteen exact directions, and the criterion is recomputed. If some move lowers it (the dark arrow), the box is not held; if no move lowers it, the box is *held*.

Held. A point x is a local minimum of E if $E(y) \geq E(x)$ for every y within some distance of x . Two of the base terms are not differentiable, crossings being a count and overlap having corners where boxes touch, so the test uses no gradient. Fix a radius d and a set V of sixteen equally spaced unit directions. For node i and direction v let $x + dv^{(i)}$ be the layout with node i moved by d along v and every other node fixed (Figure 1). Node i is *held* at radius d if

$$E(x + dv^{(i)}) \geq E(x) \quad \text{for every } v \in V, \quad (2)$$

and the estimand is

$$q_d(x) = \frac{1}{N} \#\{i : \text{node } i \text{ is held at radius } d\}. \quad (3)$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d ; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V . A person in a diagram editor moves one box at a time, so single-node moves are the editor’s own move set, and equation (2) compares two energies, so it is defined for every term. Evaluated per term it separates the criteria a layout satisfies from those it does not. The primary radius is $d = 0.02$ of the drawing width; a move that lowers the energy by less than 10^{-12} of its value is a tie. The test is deterministic: no seed, no budget. The tables report the median of q over diagrams; the fit and the weight sweep report the fraction of all boxes in a corpus held, which is lower wherever diagrams pile up at 1.

What q counts. A held fraction means different things for terms of different shape. For a smooth term near a minimum with gradient g and curvature λ along it, a node is held when $|g| < d\lambda/2$: q reads how near the minimum is. For a count (crossings) or a step function of position (gridiness, below), a node is held wherever no move of length d changes the term: q reads how much of the layout the term has stopped acting on, so the term’s value is reported beside it. For a term with a corner at exact coincidence (the alignment term A1 below), a node is held when it sits within $0.19d$ of sharing a coordinate with another node (Section 5) and is free to slide along that row or column: q counts boxes pinned in one coordinate, at a resolution the direction set fixes. Beside q the results report the median best decrease a single move finds, as a fraction of the term’s value, and the share of diagrams in which any box has an improving move.

The inverse problem. Given layouts $x^{(1)}, \dots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [11, 12]. With the directional differences $D_{iv}T_k = T_k(x + dv^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv}T_k \geq 0$ for all v , a set of linear inequalities in w . The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting makes the layout a local minimum. Two properties of that program decide what a fit can return. A term with $D_{iv}T_k < 0$ for some v at almost every node can take positive weight only where another term rises in that direction; under the base energy nothing else moves there, and the sweep of Section 4 measures how much weight such a term can carry. And a term every layout satisfies

exactly, its value zero so that $D_{iv}T_k \geq 0$ everywhere, is a zero-residual vertex on its own: the program returns all weight on it and learns nothing. The second is the degeneracy Ng and Russell [15] state for inverse reinforcement learning, where a zero reward makes every policy optimal; the repairs add a margin [16] or take demonstrations as only near-optimal [17, 11, 18]. Under any first-order fit neither rescues a term of the first kind, which carries no weight under any margin; a contrastive fit, which reads a weight off the gap between a demonstration and its alternatives, is a different estimator (Section 7).

The reference length. The length and stress terms compare distances with a reference L . Fixed at the layout’s median edge length, L is not the scale at which the term is least, so at a true minimum every node still has a scaling direction that lowers the term: **neato**’s converged stress layouts read $q = 0.07$ to 0.12 under their own criterion. L is therefore the value at which the term is least for the layout as loaded, $\sum \ell^2 / \sum \ell$ over edges for length and the same over $\|x_i - x_j\|/\delta_{ij}$ for stress, fixed before any move; the converged layouts are then held at 1.00. This is the scale-normalised stress of Smelser, Miller and Kobourov [19] and of Ahmed et al. [20], and fitting L is their repair for the pathology they report. The median and $1/\sqrt{N}$ are sensitivities (Supplement S3).

The controls. A test that can only succeed is not a test. The *converged reference*: q_d on a layout a descent has converged on must be near one, or the test lacks power for that energy. **neato**’s layouts are the reference for stress, and the end point of an uncapped descent from each hand layout is the reference for every summed energy (median q 1.00 in every corpus, Supplement S4). The *tool controls*: the same graphs with the same box sizes and edge directions laid out by **neato** (stress, **-Gstart=1**, boxes left where they overlap), **prism** (**neato** followed by its overlap removal), **dot** (layered) and ELK layered 0.12.0 [21], the engine the BPMN editors run; a check in the binary confirms each control is the same graphs. The *null*: a uniform random layout of the same graph in the same bounding box, and for the alignment term a random layout snapped to the same grid as the drawing.

Terms. With n nodes, m edges, box i of width w_i and height h_i , and L fitted to the layout under test:

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_S stress	mean over node pairs of $((\ x_i - x_j\ - L\delta_{ij})/(L\delta_{ij}))^2$, δ_{ij} the graph distance	distances proportional
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\Delta x , \Delta y)/(\Delta x + \Delta y)$	routes axis-aligned
T_A alignment	A1: mean over nodes of the distance to the nearest other node along whichever axis is nearer	every node in a row or column
gridiness	1 – fraction of nodes in an alignment of ≥ 3 (tolerance 0.005), HOLA’s definition [22], a step	every node aligned
T_N node-edge	squared shortfall of node to foreign route distance below $(w + h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u , over length	no edge runs back

The base energy is $T_C + T_O + T_L$ at weights $(1, 1, 1)$, three terms of the family that runs from Davidson and Harel’s simulated annealing [2] to the gradient-descent layouts of Ahmed et al. [23]; stress [24, 25], which force-directed tools and **neato** minimise, is the second base energy.

T_R , T_A and T_N are the candidates for what the base energies miss. A1 has a corner at exact alignment; a smooth Gaussian form, A3 with width 0.02, is the sensitivity. The reading direction u is the axis direction under which T_F is least for the layout as loaded, fixed before any move, the same rule as for L .

Routes. An edge is the polyline the file stores, attachment points and interior waypoints, or the chord between the centres where the file stores nothing. Crossings, orthogonality and node–edge separation are evaluated on the route, length and stress on the centres; when a node moves its attachment points move with it and the waypoints stay. A waypoint is stored on 0.29, 0.07 and 0.33 of the edges by corpus, and only the BPMN routes are orthogonal (0.88 axis-aligned at every segment, against 0.01 in the biological formats). The chords are the sensitivity, and no verdict moves under them.

Fitting. Weights on the simplex, the linear program above over every node and direction of a corpus (about 110,000 inequalities, HiGHS), the residual its value per node. Because the residual depends on the terms’ scales and q does not, q under the fitted weights is reported beside it, on the fitting diagrams and held out under 5×2 cross-validation (five seeded halvings, each half fitted and scored on the other). The sweep, q as the weight on one term grows from 0 with the rest on crossings and overlap, is the test of whether any positive weight on that term can be carried.

Statistics. The unit of inference is the diagram. Every median carries a 95% bootstrap interval over diagrams (1,000 resamples); an interval of [0.00, 0.00] or [1.00, 1.00] is a pileup at the boundary, and the share of diagrams at exactly that value is given where it matters. Hand against tool on one criterion is a Hodges–Lehmann shift with its distribution-free interval (Supplement S5); equivalence claims are read off the sweep, never off a failure to reject. The tests are exploratory: the plan’s hypotheses were fixed after a pilot (Supplement S8). The test and the terms are C in the repository; the descents run through `cjitter`, a library of seeded searches with uniform sampling at the same budget as its built-in control.

3 The corpora

Three collections of diagrams whose node coordinates a person chose, from three communities with different authoring tools, all redistributable, and two collections of drawings made for their own sake.

corpus	made in	in the archive	15–40 nodes	used	median m/n
WikiPathways GPML, <i>Homo sapiens</i> [26]	PathVisio	1,016	305	305	1.06
Reactome SBGN-ML [27]	curation tool	1,382	248	248	1.04
BPMN Academic Initiative [28, 29]	Signavio	29,810	6,723	300	1.11
HOLA formative drawings [22]	study tool	136	(4–13)	136	
GD Collection [14]	mixed	4,890	1,072	1,072	

The first three are working documents: Reactome diagrams are laid out by curators, WikiPathways pathways are drawn in PathVisio, and the BPMN models are students’ work in Signavio. Nobody was asked to draw nicely, and the three tools share no layout code. A diagram enters as its largest connected component of 15 to 40 nodes, container glyphs (SBGN compartments, BPMN pools and lanes) not being nodes and SBGN process glyphs being nodes; each layout is rescaled uniformly so that its bounding box fits the unit square, box sizes with it. Of the BPMN archive’s 29,810 files, 21,784 parse as models with ten or more nodes, and 13,435 of those have every edge a BPMN flow or association and the rest are EPC, Petri net and UML

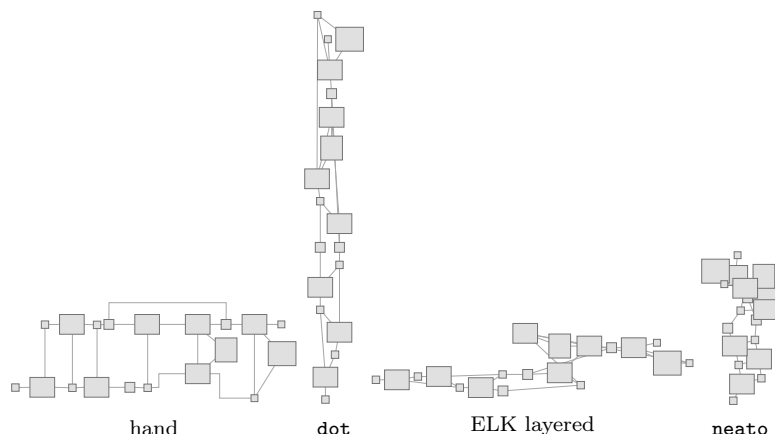


Figure 2: The same graph, the same box sizes, four layouts, one scale: the hand drawing as stored, and the tool controls of Section 2. The hand drawing and the two layered tools share the profile the test reads: held under overlap, free under length and stress. **neato** minimises stress and is the control that gives the test its power, held at 1.00 under stress where every hand layout reads 0.00; it is run as shipped, boxes left where they overlap, and its overlap-removed variant is the **prism** control of the tables. The tool layouts carry no waypoints, so their edges are drawn as chords.

models; 6,723 of the 13,435 are in the band, and the corpus is a seeded random 300 of them. The first 300 by model id, the sample the study was planned on, sits at the 99th percentile of random samples on alignment and is retired to Supplement S6. Two WikiPathways files carry coordinates converted from Reactome [30] and are dropped. Deduplication by coordinates and by structure finds three pairs in the biological corpora and none in BPMN, and removing them moves no median; the 305 WikiPathways pathways carry 153 first authors, and the alignment median’s interval clustered by author, [0.48, 0.55], is within 0.01 of the interval over diagrams (**results/dedup.py**, **results/drawers.py**).

The two further collections are drawings, not working diagrams. HOLA’s formative corpus is 17 participants each laying out the same 8 abstract graphs in the study’s own orthogonal editor, 136 drawings of 4 to 13 nodes, scraped from the study’s published SVGs; the corpus does not record whether that editor drew alignment aids, and at 4 to 13 boxes the test radius is a smaller fraction of a box than in the band (its distance-term zeros stand at radii 0.01 and 0.05 too). The GD Collection is 4,890 figures harvested from 27 editions of this field’s proceedings, 1,072 of them with a largest component in the band; who or what made each figure is not recorded, and many are the output of the algorithms the papers present. Its authors’ audit reproduces 69% of figures exactly, edge directions and box sizes are absent, so only the terms needing neither are run on it.

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 17% are integers in WikiPathways, 67% in Reactome and 63% in BPMN; the fraction of boxes sharing an exact x with another box is 0.37, 0.06 and 0.41, an exact y 0.29, 0.12 and 0.77, and either 0.54, 0.17 and 0.87 (**results/ties.py**), the figures the alignment row of Section 4 should be read against. In WikiPathways exact alignment is twice as common as integer coordinates, so it is not a rounding grid’s act. PathVisio and the BPMN editors draw a snap-to-object guide on every drag; Reactome’s curation tool does not. Section 5 takes this up.

4 What the coordinates hold

Figure 2 shows one graph under the hand and three tools. The tables are from one deterministic pass over the corpora (**results/run_all.sh**; commit, versions, seeds and corpus hashes in **results/manifest.txt**); the BPMN sample runs by the same scripts with **-corpora**

BPMN : bpmnr.

Each criterion alone. Table 1; **prism**, the A3 kernel and the summed energies are in Supplement S1.

Read down a hand column. Overlap holds every box of the median diagram at value zero, as it does in every tool layout that removes overlaps; in 5 to 18 per cent of hand diagrams some box has an overlap-lowering move, the overlap being positive but too small to print. Length and stress hold no box of the median hand diagram, and 0.85 to 0.92 of diagrams sit at exactly 0.00; **neato**’s stress layouts are held at 1.00 under the same test, so the zero is the layout’s and not the test’s. The random layouts read 0.00 too, as do **dot** and ELK, which are layered and price no distance: at this radius no layout is stationary under a distance term except a minimiser’s converged output, and the hand layouts stand with the layered engines. Alignment is where the hand layout is held and **neato**’s is not. The gridiness value puts 58, 36 and 81 per cent of hand-placed boxes in an alignment of three or more, none of **neato**’s, and 88, 82 and 76 per cent of **dot**’s, since a layer is a row; the gridiness q on **neato**’s layouts is the step’s flatness, so alignment is compared on the value, with the A1 hold beside it as the count a fit needs. The A1 hold tracks the exact-tie fractions of Section 3 to within a few points, as a corner term should. Alignment does not hold the most boxes: node–edge separation holds 0.88 to 1.00 and crossings and overlap hold every box. What separates alignment from those three is that the base energies price them and do not price it.

What the zeros mean. Beside each q the table gives *dec*, the median over diagrams of the mean best decrease a single move finds as a fraction of the term’s value, and *active*, the share of diagrams in which any box has an improving move.

term	WikiPathways			Reactome			BPMN		
	q	dec	active	q	dec	active	q	dec	active
crossings	1.00	0.000	30%	1.00	0.000	20%	1.00	0.000	3%
overlap	1.00	0.000	18%	1.00	0.000	5%	1.00	0.000	8%
node–edge	0.88	0.047	64%	1.00	0.000	50%	1.00	0.000	12%
length	0.00	0.013	100%	0.00	0.013	100%	0.00	0.014	100%
stress	0.00	0.005	100%	0.00	0.006	100%	0.00	0.007	100%
orthogonality	0.19	0.012	100%	0.17	0.027	100%	0.73	0.008	97%
alignment A1	0.52	0.066	92%	0.21	0.069	100%	0.85	0.059	63%
flow	0.59	0.010	90%	0.62	0.012	96%	0.86	0.004	71%

The three terms above the rule are flat at this radius: on BPMN only 3% of diagrams offer a crossing-reducing move, so “crossings hold every box” is a statement about the radius and these near-trees (m/n 1.04 to 1.11, against 1.30 in the Rome benchmark [31]), not about the author. The five below are live everywhere, and there the 0.00 of length and stress is a real reading: every diagram has improving moves, the mean best one worth 1.3 to 1.4 per cent of the length term and 0.5 to 0.7 per cent of stress, and none was taken. One such move is below perception; the departure is cumulative. A climber capped at the test radius removes 0.28 to 0.31 of $C+O+L$ and 0.12 to 0.15 of stress from the hand layouts (Supplement S4), and on the stress value the hand layouts (0.20 to 0.22) sit beyond **neato**’s converged layouts with every box displaced by 0.10 of the width (0.12 to 0.13), five to twelve times the converged value (0.016 to 0.042) and under half the random layouts’ (0.44 to 0.51).

Reweightings. The linear program is degenerate as Section 2 says it must be. Offered the base energy it puts all weight on the term every layout satisfies, overlap in the biological corpora and crossings in BPMN (residual per node 0.0001, 0.0000 and 0.0005; held-out q 0.987, 0.994

and 0.992). Offered all eight terms it returns the same vertex in the biological corpora, and the held-out q on WikiPathways swings [0.235, 0.991] across splits, zero-residual vertices tying differently in each; in BPMN it puts 0.81 of the weight on alignment with held-out q 0.552 [0.479, 0.983], a vertex whose summed violation (0.0003 per node) undercuts the crossings vertex's (0.0005) by the terms' scales alone: halving the crossings column returns all weight to crossings (`results/bpmnr_fit.md`). Confined to length and stress it puts everything on stress and holds 0.003 to 0.011 of the boxes: no distance-only weighting holds a hand layout. Confined to the four terms no layout satisfies exactly (orthogonality, alignment, node-edge, flow) it is identified in every corpus: alignment carries 0.95, 0.92 and 0.88 of the weight with the held-out q stable at 0.248 [0.236, 0.260], 0.149 [0.142, 0.157] and 0.651 [0.617, 0.689], the brackets the range over the ten splits. Under the smooth A3 kernel the same fit puts its weight on flow and orthogonality (0.58 and 0.32 on WikiPathways, 0.57 and 0.25 on BPMN), all of it on node-edge in Reactome with an unstable held-out hold, and 0.04 to 0.08 on alignment: the corner is what the program responds to. The sweep, the fraction of the corpus's boxes held as a weight w on one term joins crossings and overlap at equal weight:

term added to C+O	WikiPathways		Reactome		BPMN	
	$w=0.001$	$w=1$	$w=0.001$	$w=1$	$w=0.001$	$w=1$
none ($w = 0$)	0.94		0.97		0.99	
length	0.008	0.004	0.010	0.007	0.012	0.010
stress	0.005	0.003	0.008	0.006	0.013	0.011
alignment A1	0.506	0.516	0.243	0.238	0.790	0.794
flow	0.576	0.606	0.613	0.625	0.811	0.819

The fall is a step, not a slope: whatever weight length or stress carries, the sum holds under 0.015 of the boxes, because at almost every box some direction lowers the distance term and nothing else moves. Alignment and flow carry weight at the fraction they are satisfied. Under A3 the alignment row reads 0.29, 0.13 and 0.39 at $w = 0.001$, a third to half of the corner's figure, the number a gradient-trained consumer should expect. Axis-only moves, which preserve every perpendicular alignment tie, still improve length and stress on the median diagram of every corpus, and a weight of 0.001 on length beside alignment collapses the median $C+O+A_1$ hold from 0.51, 0.22 and 0.85 to 0.11, 0.07 and 0.13; and under the constrained model of the layout literature, stress subject to alignment constraints [32, 33, 34], 0.88 to 0.90 of the median diagram's boxes still have a stress-lowering move among the directions that do not raise A1 (`results/constrained.py`).

The two further collections. The same test on the drawings of Section 3 that are not working diagrams gives the same profile. HOLA's 136 drawings: crossings, overlap and node-edge 1.00, length and stress 0.00, alignment A1 0.50 with a participant-clustered interval of [0.31, 0.57] (2,000 resamples) and per-participant medians from 0.14 to 0.96. The GD Collection's 1,072 figures: crossings 1.00, length and stress 0.00 and at exactly zero in 82 and 84 per cent of figures, alignment A1 0.87. The collection's profile is also the layered tools' profile, so it cannot say whether hands or algorithms made its figures; what it says is that the corpus the standing proposal would fit first is held by the term the base energies omit and by no distance term. In the 41 to 100 node band the verdicts stand and the alignment hold falls to 0.39, 0.16 and 0.74 (Supplement S7).

5 What else could produce this

The editor snapped to a grid, so alignment is the tool's. Excluded. The detected pitch is one unit in Reactome and BPMN and none in WikiPathways, where exact alignment is twice as

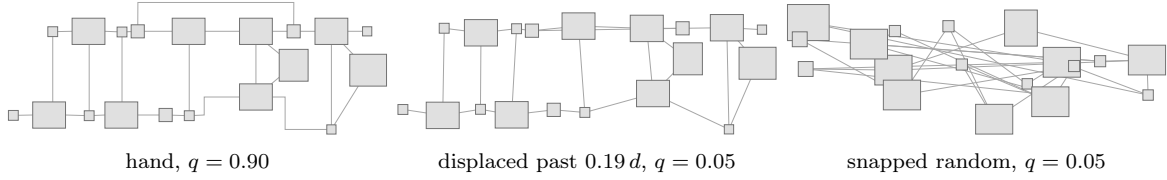


Figure 3: The grid alternative of Section 5: the grid alone does not produce the alignment hold. One BPMN model, 19 boxes; the corpus medians of the three panels are 0.85, 0.19 and 0.07. Left, the drawing as its author left it. Centre, every box displaced by up to 15 units, three times the $0.19d$ past which the directional test resolves a misalignment. Right, every box placed uniformly at random in the same bounding box and snapped to the same grid: the grid is available and the author is not. Box sizes are unchanged throughout. The hand panel draws the routes the file stores; the two control panels draw chords, their boxes no longer being where the routes ran.

common as integer coordinates. A layout snapped to the same grid, with the same boxes and the same bounding box and no author, holds 0.06 to 0.09 over three seeds in every corpus (Figure 3); snapped to the plausible working pitch of five units, 0.076 (first 300). The grid is available to the random layout and does not produce the effect. Displacing every hand-placed box by up to half a pitch leaves the hold within 0.02 of where it was, and that test cannot fail, for a reason in the instrument: a box offset by ϵ from its nearest agreement is released only by a move whose component toward the agreement, a , is less than 2ϵ , since the move lands at $|\epsilon - a|$; the smallest such component among sixteen directions is $0.38d$, so release begins at $\epsilon > 0.19d$, which is 3 to 5 units against a half pitch of 0.5. Displacing past that takes the hand layouts to the snapped random level (0.06 at a half-width of 120 units; Supplement S7).

The editor snapped to the neighbours, so alignment is the guide’s. Not separable on these corpora, and conceded: a grid control cannot see snap-to-object guides, and the hold orders as the editors’ guides of Section 3 do, highest where guides meet every drag and lowest where they are absent. Two pieces of evidence say no single editor’s aid produces it: HOLA’s drawings come from a study tool and hold 0.50, and WikiPathways is flat at 0.47 to 0.53 across four generations of pathway numbers, so no gradient follows PathVisio’s releases (`results/era.md`). Separating hand from guide needs a guide-free editor or an interaction log, and these corpora carry neither. Marriott et al. [35] bound the other side: in a recall task node alignment was the one feature participants did not preserve, so the alignment these coordinates hold is a property of how the layouts were made, not evidence that anyone perceived it. Every claim here is about the coordinates as data for a fit, and for a fit the attribution changes nothing: the coordinates hold a term the energy lacks either way.

The lanes hold the boxes. 82% of BPMN boxes sit in a lane or pool, and a horizontal lane constrains y . Of the held boxes, 28% are held by a column, which no horizontal lane imposes, 14% by a row shared across lanes, and 58% by a row within one lane, and those lanes are a median 6.3 box heights tall, so the row inside them was chosen (`results/lanes.py`).

The rest, each with the test that excluded it:

6 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [1] and were made an explicit energy by Davidson and Harel [2]. Purchase [3] gave the criteria their metrics, among them node orthogonality, the fit of node positions to an imaginary grid; Bennett et al.’s survey [36] lists it and uniform edge length side by side, each with a Gestalt rationale and, as of

alternative	test	result
Centre distances on box diagrams	border-to-border distances, L refitted	hand 0.00 in every corpus; neato , which minimised the centre form, not held under the border form (Supplement S3)
The radius and the directions	$d \in \{0.005, 0.01, 0.02, 0.05\}$; 8 to 64 directions	hand verdicts the same at every d and count; the A1 level falls with the direction count and the gap to the control does not (Supplement S2)
Near-trees make overlap and crossings easy	q against m/n over diagrams (results/sparsity.md)	crossings track density by hand (Spearman -0.05 to -0.37) and under dot (-0.30 to -0.48) alike; length and stress are 0.00 in the sparse and the dense half both
Reading direction is the missing constraint	the flow term	flow holds 0.59 to 0.86 of hand-placed boxes where dot and ELK hold 1.00: a criterion partly satisfied, the person tolerating a backward edge
The tools are the wrong tools	ELK layered, the engine the BPMN editors run	its profile is dot 's: distance terms 0.00, flow 1.00, alignment 0.42 to 0.59

2007, neither evaluated. The test here is such an evaluation, at the coordinates, and it separates the two.

work	what it gives, and what it leaves
<i>By having people draw</i>	
van Ham, Rogowitz [37]	71 of 73 hand layouts raise edge-length variance above the force-directed optimum half of them started from
Dwyer et al. [38]	194 judges' choices among 32 user layouts and three automatic ones track stress; the user layouts carry 3 to 9 times the force-directed stress in that paper's Table 2
Purchase, Pilcher, Plimmer [39]	drawing from adjacency lists: crossing removal the strongest aesthetic, alignment to a grid important
Kieffer et al. [22]	17 participants' layouts of 8 graphs ranked by 66 judges: gridiness correlates with rank on all 8, the source of HOLA's alignment step
Marriott et al. [35]	25 participants redrawing small diagrams from memory preserve symmetry, collinearity and orthogonality, and not node alignment
Mooney et al. [40]	people prefer low stress and find shortest paths better as it falls, graphs to 50 nodes
Mooney, Hegemann, Wolff, Wybrow, Purchase [14]	4,890 figures harvested from these proceedings, provenance not per figure; measured in Section 4
<i>By learning from examples</i>	
Masui [5]; Spönemann et al. [6]; Klammler et al. [7]; Zhang et al. [9]	weights from good and bad layouts, from selections, from good-against-worsened pairs, from 64,222 preference labels; none tests whether an example is a minimum of the learned energy, none fits coordinates
Flud [8]	annealing continued from crowd layouts of pathway diagrams, the closest prior descent from a hand layout, without a cap, a reference or a per-term reading
O'Donovan, Agarwala, Hertzmann [13]	a 122-parameter energy over single-page designs fitted by inverse optimisation, each example taken as optimal: the assumption examined here; their term set contains alignment

Alignment as a human criterion is known since 2012 in graph drawing [39] and since 1985 in drawing editors: Pavlidis and Van Wyk's beautifier inferred alignment constraints from the near-alignments of a rough drawing [41], Igarashi et al. made the inference interactive [42], and the guides of Section 5 are their descendants. Energies and constraint systems that price it

exist: HOLA’s gridiness step [22], O’Donovan’s alignment terms [13], and stress under alignment constraints [32, 33, 34]. The new part is the count of how many hand-placed boxes it holds beside the counts for the criteria the base energies weight, and the step under the distance terms. Van Wageningen, Mchedlidze and Telea [43] show drawings with identical metric values that differ in quality, so a fitted energy with no residual would still not be the energy people use; the claims here stay on the negative side. The inverse problem and its degeneracy are Section 2’s, and Marriott et al.’s test of feature preservation against random redrawings [35] is the nearest precedent in layout to the nulls used here.

7 What follows

Coordinates a person accepted are not minima of the energy in use. The failure is specific: overlap is met exactly, uniform edge length and stress are nowhere near met, and what holds the boxes instead is an alignment into rows and columns that neither base energy prices. The charge is bounded twice: dot and ELK hold alignment at 0.42 to 1.00 without pricing it, since layering aligns as a side effect, so the missing term convicts the energies and not the layered engines; and the energies that do price alignment, HOLA’s, O’Donovan’s and the constraint systems, are not the ones a coordinate fit to a graph-drawing corpus would be offered.

For a tool that learns its weights. Offer alignment among the terms or the fit returns nothing the drawing did. That is an addition, not a subtraction: edge length and stress keep their weight, since a term inert at human coordinates is not thereby unwanted, and the preference evidence runs the other way [40, 38]. People prefer the layouts a stress minimiser makes, do not draw that way, and do not notice the alignment they do draw [35]; no study yet takes production and perception from one corpus. An energy fitted to coordinates a snap guide produced learns the guide: an editor that already snaps gains a term it enforces on every drag, and a layout engine without a guide gains what the guide gives. The number to report for a fitted energy is the fraction of a held-out drawing’s boxes it holds under the terms no layout satisfies exactly, $q_{0.02}$ at sixteen directions with the fitted L , with the decrease and active columns beside it, since a bare held fraction is bought by weighting a term everyone satisfies: the degenerate fit reads 0.99. The gain is small and measured: replacing each box in turn against the frozen rest, alignment does not move the median distance to the drawing’s position and raises the share of boxes landing on the drawing’s row or column by a fifth to a half (`results/reinsert.md`).

Where the test applies. Before any decomposable objective is fitted to artefacts taken as optimal, wherever the analyst can move one variable at a time, the per-term held fraction with a converged reference and a random null sorts the terms into satisfied (held at value zero, the source of the trivial solution), inert (flat at the demonstration radius) and violated (an improving move at almost every demonstration, so no first-order fit can weight it), and a candidate term that holds the demonstrations while the fitted energy holds none is a missing feature found without fitting. It does not transfer to sequential decisions, whose condition is optimality along a trajectory. A contrastive fit reads a different quantity: hand layouts carry half the stress of random layouts of the same graphs (0.20 to 0.22 against 0.44 to 0.51) and more length non-uniformity (0.22 to 0.28 against 0.19 to 0.22, `results/nulls.md`), so a preference or likelihood fit would weight stress and not length uniformity, where the coordinate fit weights neither.

Limits. The alignment count is specific to box-and-arrow diagrams of 4 to 100 boxes drawn in editors with guides, it falls with size, and its level is stated at sixteen directions, where the corner’s resolution is $0.19d$; at sixty-four it reads 0.39, 0.07 and 0.82 against a chance line of 0.03, 0.03 and 0.00. The hand is not separated from the guide. The GD Collection is measured on chord edges with no box sizes and does not record who made its figures. The distance terms

read 0.00 on every layout that is not a minimiser’s output, so the negative result separates hand layouts from converged ones and from nothing else. The BPMN corpus is a sample of 300 from 6,723. The first-order residual is a scale-dependent number and is reported only beside q .

8 Reproducibility

Parsers, the energy, the directional test, the descent, the fitting scripts, the parsed corpora and every table’s script are in the repository under `example/diagrams`, each analysis beside its committed output, and `results/manifest.txt` records the commit, the tool versions, the seeds and the SHA-256 of the raw archives and the corpus files. The directional test is deterministic and the descents are deterministic from their seed. The plan the study was run to is beside the paper with its outcomes in Supplement S8, and the supplement is `stationary_supplement.pdf` beside this file.

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